WHEN IS THE DIVISOR SUBALGEBRA OF A PARTIAL FLAG VARIETY A POINCARÉ-SYMMETRIC ALGEBRA?

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ABSTRACT. Let $X = \mathrm{Fl}(a_1, \dots, a_m; \mathbb{C}^n)$ be a type A partial flag variety of complex dimension d, and let $L \subseteq H^*(X; \mathbb{Q})$ be the subalgebra generated by $H^2(X; \mathbb{Q})$. We determine exactly when the Hilbert function of L is symmetric, i.e. when $\dim L^k = \dim L^{d-k}$ for all k. This holds if and only if X is a Grassmannian ($m = 2$) or at most one of the block sizes a_j exceeds one. In every other case the symmetry already fails in degree two: $\dim L^{d-2} > \dim L^2$. The proof combines hard Lefschetz and the Hodge–Riemann relations with an explicit computation of degree-four intersection numbers through the volume polynomial of a coadjoint orbit and the orthogonality of Schur functions under the apolar pairing. These reduce the problem to a single 4×4 determinant in three variables. We also report exhaustive computations for all partial flag varieties with $n \leq 8$.

1. INTRODUCTION

Let X be a smooth complex projective variety of complex dimension d and let $L = L(X) \subseteq H^*(X; \mathbb{Q})$ be the subalgebra generated by divisor classes, $L = \mathbb{Q}[H^2(X; \mathbb{Q})]$, graded by $L^k = L \cap H^{2k}(X; \mathbb{Q})$. When $H^*(X)$ is generated by H^2 (for instance for toric varieties or complete flag varieties), $L = H^*(X)$ satisfies Poincaré duality, and in particular its Hilbert function is symmetric: $\dim L^k = \dim L^{d-k}$. In general L is a proper subalgebra and need not satisfy any duality. Nevertheless, an ample class $\omega \in L^1$ and the hard Lefschetz theorem give injections

$$\omega^{d-2k} \colon L^k \hookrightarrow L^{d-k} \qquad (0 \leq k \leq d/2), \tag{1.1}$$

so that $\dim L^k \leq \dim L^{d-k}$ always, and one may ask when equality holds throughout. Subalgebras generated by divisors, and their Lefschetz and Hodge-theoretic properties, play a central role in combinatorial Hodge theory; see e.g. [1, 8].

For partial flag varieties this question was recorded as open in a problem list on the work of June Huh [4], with the observation that symmetry fails for $\mathrm{Fl}(2, 3; \mathbb{C}^5)$. The purpose of this note is to answer it completely in type A.

Notation. A composition $\mathbf{a} = (a_1, \dots, a_m)$ of n with $m \geq 2$ parts determines the partial flag variety

$$X_{\mathbf{a}} = \mathrm{Fl}(a_1, a_1 + a_2, \dots, a_1 + \cdots + a_{m-1}; \mathbb{C}^n)$$

of flags $0 = V_0 \subset V_1 \subset \cdots \subset V_m = \mathbb{C}^n$ with $\dim V_j/V_{j-1} = a_j$. Its complex dimension is $d = \sum_{j<l} a_j a_l$. We write $b(\mathbf{a}) = \#\{j : a_j \geq 2\}$ for the number of *big blocks*.

Theorem 1.1. *Let $\mathbf{a} = (a_1, \dots, a_m)$, $m \geq 2$, and let $L \subseteq H^*(X_{\mathbf{a}}; \mathbb{Q})$ be the subalgebra generated by H^2. The following are equivalent:*

(i) $\dim L^k = \dim L^{d-k}$ *for all k;*

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(ii) $m = 2$, *or* $b(\mathbf{a}) \le 1$.

Moreover, if (ii) *fails, then* $\dim L^{d-2} > \dim L^2$.

Thus symmetry holds exactly for Grassmannians and for those partial flag varieties in which all blocks but one have size one (these include complete flag varieties and, e.g., $\mathrm{Fl}(1,2;\mathbb{C}^n)$), and in all other cases it fails in the lowest possible degree. The smallest non-symmetric examples are the three orderings of the block sizes $(2,2,1)$; for $\mathbf{a} = (2,1,2)$, i.e. $\mathrm{Fl}(2,3;\mathbb{C}^5)$, the Hilbert function of L is

$$(1,2,3,4,5,5,4,2,1),$$

which is not symmetric about $d/2 = 4$. Note that the answer depends only on the multiset of block sizes, not on their order.

Strategy of the proof. The implication (ii)⇒(i) is elementary: if $b \le 1$ the Whitney sum formula shows that H^* is generated by H^2 (Section 3). For the converse we show $\dim L^{d-2} > \dim L^2$. By the Hodge–Riemann relations, $\dim L^2$ equals the dimension of the span Φ of the second partial derivatives of the volume polynomial

$$V(\mu) = \int_X \frac{(\mu_1 y_1 + \cdots + \mu_m y_m)^d}{d!} = \kappa \prod_{j<l} (\mu_j - \mu_l)^{a_j a_l},$$

while $\dim L^{d-2}$ is the dimension of the span of the polynomials $\mu \mapsto \int_X c\,\omega_\mu^{d-2}$ for $c \in H^4$ (Section 4). Splitting a big block and using the orthogonality of Schur functions under the apolar pairing, we compute this polynomial for $c = p_2(x_{B_k})$, the second power sum of the Chern roots of a big block. It turns out to lie outside Φ exactly when a certain explicit rational function lies outside an explicit span (Section 5). A limiting argument reduces the latter to configurations with three blocks (Section 6), where it becomes the non-vanishing of a single determinant,

$$4A^2B^3C^3(B-1)(B+1)(AC+BC-1),$$

in the block sizes A, B, C (Section 7).

Computations. We computed the Hilbert function of L for all 246 compositions with $3 \le n \le 8$ and $m \ge 2$ in Macaulay2 [9]. The results (Table 1) agree with Theorem 1.1, and in each of the 148 non-symmetric cases the asymmetry is visible in degree two.

n	3	4	5	6	7	8
compositions with $m \ge 2$	3	7	15	31	63	127
symmetric Hilbert function of L	3	7	12	18	25	33
non-symmetric	0	0	3	13	38	94

Table 1. Exhaustive computation for $n \le 8$, in agreement with Theorem 1.1.

2. Preliminaries

2.1. **The cohomology ring.** Let $B_1, \dots, B_m$ be the consecutive blocks of $\{1, \dots, n\}$ of sizes $a_1, \dots, a_m$, and let $W_{\mathbf{a}} = S_{a_1} \times \cdots \times S_{a_m}$. By Borel's theorem [3] (see also [6]),

$$H^*(X_{\mathbf{a}};\mathbb{Q}) \;\cong\; \mathbb{Q}[x_1, \dots, x_n]^{W_{\mathbf{a}}} \big/ \big(e_1(x), \dots, e_n(x)\big),$$

with $\deg x_i = 2$, where x_i ($i \in B_j$) are the Chern roots of the tautological bundle V_j/V_{j-1}, up to a global sign convention that plays no role below. For a block B and a symmetric polynomial f we write $f(B) = f(x_i : i \in B)$. In particular

$$y_j := e_1(B_j) = \sum_{i \in B_j} x_i, \qquad y_1 + \cdots + y_m = 0,$$

and $y_1, \dots, y_{m-1}$ form a basis of $H^2(X;\mathbb{Q})$. The algebra L is the image of $\mathbb{Q}[y_1, \dots, y_m]$. For $\mu \in \mathbb{Q}^m$ we put $\omega_\mu = \sum_j \mu_j y_j$; the ample cone consists of the ω_μ with μ strictly monotone (in the appropriate direction).

2.2. **Lefschetz package.** X is a smooth projective variety, so for an ample class ω the hard Lefschetz theorem and the Hodge–Riemann bilinear relations hold [7, 11]. In degree four this means the following. Write $P^k \subseteq H^{2k}$ for the primitive subspaces with respect to ω. Then

$$H^4 = \omega^2 H^0 \oplus \omega P^1 \oplus P^2,$$

the summands are orthogonal for the form $Q_\omega(u,v) = \int_X \omega^{d-4} uv$, and Q_ω is definite on each summand (with signs $+, -, +$ respectively).

2.3. **The polynomials P_c.** For $c \in H^4(X;\mathbb{Q})$ define the polynomial

$$P_c(\mu) = \int_X c \cdot \frac{\omega_\mu^{d-2}}{(d-2)!} = \sum_{|\beta| = d-2} \frac{\mu^\beta}{\beta!} \int_X c\, y^\beta, \qquad V(\mu) = \int_X \frac{\omega_\mu^d}{d!}.$$

Since $\partial_{\mu_i} \omega_\mu^N / N! = y_i\, \omega_\mu^{N-1}/(N-1)!$, we have

$$P_{y_i y_j} = \partial_i \partial_j V, \qquad \partial_i := \partial/\partial\mu_i. \tag{2.1}$$

The volume polynomial is the symplectic volume of a coadjoint orbit of $U(n)$ and is given by the classical product formula (Harish-Chandra, Duistermaat–Heckman; see [5, 2])

$$V(\mu) = \kappa_{\mathbf{a}} \prod_{1 \le j < l \le m} (\mu_j - \mu_l)^{a_j a_l}, \qquad \kappa_{\mathbf{a}} \in \mathbb{Q}^\times. \tag{2.2}$$

Changing the sign convention for the x_i replaces μ by $-\mu$; all statements below are invariant under this substitution. We will repeatedly use the rational functions

$$g_k = \sum_{l \ne k} \frac{a_l}{\mu_k - \mu_l}, \qquad h_k = \sum_{l \ne k} \frac{a_l}{(\mu_k - \mu_l)^2}.$$

Logarithmic differentiation of (2.2) gives $\partial_k \log V = a_k g_k$, $\partial_j \partial_k \log V = a_j a_k (\mu_j - \mu_k)^{-2}$ for $j \ne k$, and $\partial_k^2 \log V = -a_k h_k$. Hence

$$\frac{\partial_i \partial_j V}{V} = a_i a_j \Big(g_i g_j + \frac{1}{(\mu_i - \mu_j)^2} \Big) \ (i \ne j), \qquad \frac{\partial_k^2 V}{V} = a_k \big(a_k g_k^2 - h_k \big). \tag{2.3}$$

We let $\Phi = \operatorname{span}_{\mathbb{Q}} \{ \partial_i \partial_j V \}$ and $S = V^{-1}\Phi$. By (2.3),

$$S = \operatorname{span}_{\mathbb{Q}} \Big\{ a_i a_j \big(g_i g_j + (\mu_i - \mu_j)^{-2} \big) \ (i < j), \ \ a_k \big(a_k g_k^2 - h_k \big) \Big\},$$

a space of rational functions in μ.

3. The symmetric cases

Proposition 3.1. *If $m = 2$ or $b(\mathbf{a}) \le 1$, then $\dim L^k = \dim L^{d-k}$ for all k.*

Proof. If $m = 2$ then H^2 is spanned by y_1, and $L^k = \mathbb{Q}\, y_1^k$ for $0 \le k \le d$, because y_1 is ample up to sign and so $y_1^d \neq 0$. Hence $\dim L^k = 1$ for $0 \le k \le d$.

Suppose $b \le 1$. If $b = 0$, then X is the complete flag variety, H^* is generated by $x_1, \dots, x_n = y_1, \dots, y_n$, and $L = H^*$. If $b = 1$, let B be the unique big block. All other blocks are singletons $\{i\}$ with $x_i = y_{j(i)} \in L$. In $H^*(X)[t]$ we have $\prod_{i=1}^n (1 + x_i t) = 1$, hence

$$\sum_{r \ge 0} e_r(B) t^r = \prod_{i \in B} (1 + x_i t) = \prod_{i \notin B} (1 + x_i t)^{-1},$$

and every $e_r(B)$ is a polynomial in the singleton classes, so it lies in L. Since H^* is generated by the $e_r(B_j)$, we get $L = H^*(X)$, which satisfies Poincaré duality. □

4. Reduction to a statement about volume polynomials

Proposition 4.1. *Let $\mathcal{P} = \{P_c : c \in H^4(X; \mathbb{Q})\}$, a $\mathbb{Q}$-vector space of polynomials containing Φ. Then $\dim L^{d-2} = \dim \mathcal{P}$ and $\dim L^2 = \dim \Phi$. In particular, if $\mathcal{P} \neq \Phi$ then $\dim L^{d-2} > \dim L^2$.*

Proof. L^{d-2} is spanned by the monomials y^β, $|\beta| = d - 2$, and the Poincaré pairing $H^{2(d-2)} \times H^4 \to \mathbb{Q}$ is perfect. Hence $\dim L^{d-2}$ is the rank of the matrix $\big(\int_X y^\beta c\big)_{\beta, c}$, which is the dimension of the space of linear functionals $y^\beta \mapsto \int y^\beta c$. That space is identified with $\mathcal{P}$ through the coefficients of P_c. This proves the first equality.

The map $L^2 \to \Phi$, $c \mapsto P_c$, is surjective by (2.1). Suppose $c \in L^2$ and $P_c = 0$, so $\int c\, y^\beta = 0$ for all $|\beta| = d - 2$. Fix an ample $\omega \in L^1$ and write $c = \lambda\omega^2 + \omega\ell + p$ with $\ell \in P^1$, $p \in P^2$. Since $H^2 = L^1$, we have $\ell \in L$ and therefore $p \in L^2$. The classes $\omega^{d-4}\omega^2$, $\omega^{d-4}\omega\ell$ and $\omega^{d-4}p$ lie in L^{d-2}, so c pairs to zero with each of them. By orthogonality of the Lefschetz decomposition, $Q_\omega(c, \omega^2) = \lambda \int \omega^d = 0$, $Q_\omega(c, \omega\ell) = Q_\omega(\omega\ell, \omega\ell) = 0$ and $Q_\omega(c, p) = Q_\omega(p, p) = 0$. Definiteness gives $\lambda = 0$, $\ell = 0$ and $p = 0$. Thus $L^2 \to \Phi$ is an isomorphism.

Finally, if $\dim L^{d-2} = \dim L^2$ then $\dim \mathcal{P} = \dim \Phi$, so $\mathcal{P} = \Phi$. □

By Proposition 4.1, Theorem 1.1 follows once we find, whenever $m \ge 3$ and $b \ge 2$, a class $c \in H^4$ with $P_c \notin \Phi$. We will take $c = p_2(B_k) = \sum_{i \in B_k} x_i^2$ for a big block B_k.

5. The degree-four class of a big block

Throughout this section fix a block $B = B_k$ of size $a = a_k \ge 2$. We use the apolar pairing on homogeneous polynomials of the same degree in variables $z = (z_1, \dots, z_a)$,

$$\langle f, g \rangle = f(\partial_z)\, g \in \mathbb{Q}, \qquad \langle z^\alpha, z^\beta \rangle = \alpha!\, \delta_{\alpha\beta}.$$

Let $\Delta(z) = \prod_{s<t} (z_s - z_t)$, of degree $N = \binom{a}{2}$, and let $\delta = (a - 1, a - 2, \dots, 0)$.

Lemma 5.1. *For partitions λ, ν with at most a parts and $|\lambda| = |\nu|$, $\langle \Delta s_\lambda, \Delta s_\nu \rangle = \delta_{\lambda\nu} N_\lambda$, where $N_\lambda = a! \prod_{i=1}^a (\lambda_i + \delta_i)! > 0$. Moreover, for every symmetric polynomial f of degree 2 and every symmetric g of degree 2,*

$$\big\langle z^\delta f,\ \Delta g \big\rangle = \tfrac{1}{a!} \big\langle \Delta f,\ \Delta g \big\rangle.$$

Proof. By the bialternant formula $\Delta s_\lambda = \sum_{\sigma \in S_a} \operatorname{sgn}(\sigma) z^{\sigma(\lambda + \delta)}$ [10]. The exponents $\sigma(\lambda + \delta)$ are pairwise distinct because $\lambda + \delta$ is strictly decreasing, and they never coincide with those of Δs_ν for $\nu \neq \lambda$. The first claim follows from $\langle z^\alpha, z^\beta \rangle = \alpha! \delta_{\alpha\beta}$.

For the second, the pairing is S_a-invariant, and $G = \Delta g$ is alternating, so $\langle \sigma F, G\rangle = \operatorname{sgn}(\sigma)\langle F, G\rangle$. Averaging $F = z^\delta f$ over S_a with signs and using $\sum_\sigma \operatorname{sgn}(\sigma)\sigma(z^\delta) = \Delta$ gives the formula. □

Proposition 5.2. *There is a constant $\gamma_a \neq 0$, depending only on a, such that*

$$P_{p_2(B_k)} \equiv \gamma_a\, a\, V\,(g_k^2 - h_k) \pmod{\Phi}.$$

Consequently $P_{p_2(B_k)} \in \Phi$ if and only if $g_k^2 \in S$.

Proof. Let X' be the partial flag variety obtained by splitting the block B into a singletons, and let $\pi\colon X' \to X$ be the forgetful map. It is a fiber bundle with fiber the complete flag variety $\mathrm{Fl}(\mathbb{C}^a)$, and on X' the Chern roots x_s, $s \in B$, are honest classes. Fix the staircase class $\Theta = \prod_{s=1}^{a} x_{b+s}^{a-s}$, where $B = \{b+1, \dots, b+a\}$. Then $\pi_*(\Theta) = \varepsilon$ with $\varepsilon = \pm 1$, since Θ restricts to a generator of the top cohomology of the fiber.

Introduce parameters $\lambda_i = \mu_j$ for $i \in B_j$, $j \neq k$, and $\lambda_{b+s} = \mu_k + z_s$. Since ∂_{z_s} applied to $e^{\sum \lambda_i x_i}$ multiplies by x_{b+s}, for any symmetric f in a variables the projection formula gives

$$\int_X f(x_B)\, e^{\omega_\mu} = \varepsilon \int_{X'} f(x_B)\, \Theta\, e^{\sum_i \lambda_i x_i} = \varepsilon \Big[f(\partial_z)\, z^\delta(\partial_z)\, V'(\mu, z) \Big]_{z=0},$$

where V' is the volume polynomial of X'. By the product formula for X',

$$V'(\mu, z) = \kappa'\, \Delta(z)\, \mathrm{Cr}(\mu, z), \qquad \mathrm{Cr}(\mu, z) = \prod_{s=1}^{a} \prod_{l \neq k} (\mu_k + z_s - \mu_l)^{a_l} \prod_{\substack{j<l \\ j,l \neq k}} (\mu_j - \mu_l)^{a_j a_l}.$$

Note that Cr is symmetric in z and $\mathrm{Cr}(\mu, 0) = \prod_{j<l} (\mu_j - \mu_l)^{a_j a_l}$. Expand

$$\mathrm{Cr}(\mu, z) = F_0 + F_1\, p_1(z) + F_2'\, p_1(z)^2 + F_2''\, p_2(z) + O(|z|^3),$$

with coefficients $F_\bullet$ depending on μ. The operator $f(\partial_z) z^\delta(\partial_z)$ has order $N + \deg f$. Applied to $\Delta(z)\, G(z)$ with G homogeneous of degree e and evaluated at $z = 0$, it vanishes unless $e = \deg f$.

Taking $f = 1$ gives $V = \varepsilon\kappa' \langle z^\delta, \Delta\rangle\, \mathrm{Cr}(\mu, 0)$, consistent with (2.2). Taking $f = p_2$ and using Lemma 5.1 with $p_2 = s_{(2)} - s_{(1,1)}$ and $p_1^2 = s_{(2)} + s_{(1,1)}$ gives

$$P_{p_2(B)} = \varepsilon\kappa' \big(c' F_2' + c'' F_2'' \big), \qquad c' = \frac{N_{(2)} - N_{(1,1)}}{a!}, \quad c'' = \frac{N_{(2)} + N_{(1,1)}}{a!};$$

here we use that P_c is the homogeneous component of degree $d-2$ of $\mu \mapsto \int_X c\, e^{\omega_\mu}$, and only the quadratic Taylor terms of Cr contribute.

Now consider the two second-order quantities

$$\mathcal{D} = \sum_{s=1}^{a} \partial_{z_s}^2 \mathrm{Cr}\Big|_{z=0} = 2a\,(F_2' + F_2''), \qquad \mathcal{T} = \Big(\sum_s \partial_{z_s} \Big)^2 \mathrm{Cr}\Big|_{z=0} = 2a^2 F_2' + 2a F_2''.$$

Since Cr depends on μ_k and z only through the sums $\mu_k + z_s$, we have $\mathcal{T} = \partial_k^2\, \mathrm{Cr}(\mu, 0)$, a nonzero multiple of $\partial_k^2 V \in \Phi$. Solving for F_2', F_2'' gives

$$c' F_2' + c'' F_2'' = \frac{a\, c'' - c'}{2a(a-1)}\, \mathcal{D} + \frac{c' - c''}{2a(a-1)}\, \mathcal{T}, \qquad a\, c'' - c' = \frac{(a-1)N_{(2)} + (a+1)N_{(1,1)}}{a!} > 0.$$

Finally, logarithmic differentiation in z_s at $z = 0$ gives $\partial_{z_s} \log \mathrm{Cr} = g_k$ and $\partial_{z_s}^2 \log \mathrm{Cr} = -h_k$, so $\partial_{z_s}^2 \mathrm{Cr} = \mathrm{Cr}\,(g_k^2 - h_k)$ and $\mathcal{D} = a\, \mathrm{Cr}(\mu, 0)\,(g_k^2 - h_k)$. This proves the congruence with $\gamma_a \neq 0$.

For the last assertion, (2.3) gives

$$a\, V (g_k^2 - h_k) - \partial_k^2 V = -a(a-1)\, V\, g_k^2.$$

Since $a \geq 2$ and $\partial_k^2 V \in \Phi$, we get $P_{p_2(B_k)} \in \Phi \iff V g_k^2 \in \Phi \iff g_k^2 \in S$. □

6. Reduction to three blocks

To emphasize the dependence on the composition we write $S(\mathbf{a})$, $g_k^{(\mathbf{a})}$, and so on. These are rational functions of $\mu_1, \dots, \mu_m$.

Lemma 6.1. *Let $m \geq 3$, $1 \leq k \leq m$ and $r \neq k$. Let $\mathbf{a}'$ be the composition obtained by deleting a_r, with variables μ_l, $l \neq r$. If $(g_k^{(\mathbf{a})})^2 \in S(\mathbf{a})$, then $(g_k^{(\mathbf{a}')})^2 \in S(\mathbf{a}')$.*

Proof. Write $(g_k^{(\mathbf{a})})^2 = \sum_\gamma c_\gamma G_\gamma$ with constants c_γ and G_γ the spanning functions of $S(\mathbf{a})$. Fix generic values of μ_l, $l \neq r$, and let $\mu_r \to \infty$. Then $a_r/(\mu_l - \mu_r) \to 0$, so $g_l^{(\mathbf{a})} \to g_l^{(\mathbf{a}')}$ and $h_l^{(\mathbf{a})} \to h_l^{(\mathbf{a}')}$ for $l \neq r$, while $g_r^{(\mathbf{a})}, h_r^{(\mathbf{a})}, (\mu_l - \mu_r)^{-2} \to 0$. Hence every spanning function involving the index r tends to 0, and every other one tends to the corresponding spanning function of $S(\mathbf{a}')$. Passing to the limit in the identity proves the claim. □

If $m \geq 3$ and $b \geq 2$, choose big blocks $k \neq l$ and any third block r. Applying Lemma 6.1 repeatedly to delete all other blocks, we see that it suffices to prove $g_1^2 \notin S$ for every three-block composition (A, B, C) with $A, B \geq 2$ and $C \geq 1$. Here S and g_1 are formed with $\mathbf{a} = (A, B, C)$, and the order of the blocks is irrelevant because all functions involved are symmetric under relabeling.

7. Three blocks

Proposition 7.1. *Let $\mathbf{a} = (A, B, C)$ with $A, B \geq 2$ and $C \geq 1$. Then $g_1^2 \notin S(\mathbf{a})$.*

Proof. All functions in S, as well as g_1^2, are invariant under translation $\mu \mapsto \mu + c(1,1,1)$ and homogeneous of degree -2. Hence a linear relation among them holds identically if and only if it holds on the slice $\mu = (0, 1, t)$. On this slice, multiply by $t^2(t-1)^2$. The six spanning functions G_{12}, G_{13}, G_{23} (from $a_i a_j (g_i g_j + (\mu_i - \mu_j)^{-2})$) and G_{11}, G_{22}, G_{33} (from $a_k(a_k g_k^2 - h_k)$), together with g_1^2, become polynomials in t of degree at most four:

$$
\begin{aligned}
G_{12} &= -ABt(t-1)\big(ABt^2 - ABt + ACt - AC - BCt - C^2 - t^2 + t\big),\\
G_{13} &= -AC(t-1)\big(ABt^2 - ABt + ACt - AC + B^2t^2 + BCt - t + 1\big),\\
G_{23} &= BCt\big(A^2t^2 - 2A^2t + A^2 + ABt^2 - ABt - ACt + AC - BCt + t\big),\\
G_{11} &= A(t-1)^2\big(AB^2t^2 + 2ABCt + AC^2 - Bt^2 - C\big),\\
G_{22} &= Bt^2\big(A^2Bt^2 - 2A^2Bt + A^2B - 2ABCt + 2ABC - At^2 + 2At - A + BC^2 - C\big),\\
G_{33} &= C\big(A^2Ct^2 - 2A^2Ct + A^2C + 2ABCt^2 - 2ABCt - At^2 + 2At - A + B^2Ct^2 - Bt^2\big),\\
g_1^2 &= (t-1)^2(Bt + C)^2.
\end{aligned}
$$

Consider the 5×7 matrix M of coefficients of $1, t, \dots, t^4$. Its first six columns span a space of dimension at most 3: the generic rank is 3, and the rank can only drop under specialization. The 4×4 minor of M on the rows t^1, t^2, t^3, t^4 and the columns $G_{12}, G_{13}, G_{23}, g_1^2$ equals

$$4A^2B^3C^3(B-1)(B+1)(AC + BC - 1),$$

which is nonzero for $A, B \geq 2$, $C \geq 1$. Hence adjoining the column of g_1^2 raises the rank to at least 4, so $g_1^2 \notin S$. □

Remark 7.2. The factor $B-1$ is essential. When $B=1$ the minor vanishes, consistent with Proposition 3.1: for $b=1$ the class p_2 of the unique big block lies in L, and indeed then $g_k^2 \in S$ for the big block k. The rank statement and the determinant are elementary polynomial identities and can be checked by hand or with any computer algebra system. A script is available from the author.

8. Proof of Theorem 1.1

(ii)⇒(i) is Proposition 3.1. Conversely, suppose $m \geq 3$ and $b \geq 2$, and let B_k be a big block. By Lemma 6.1 and Proposition 7.1, $g_k^2 \notin S(\mathbf{a})$. By Proposition 5.2, $P_{p_2(B_k)} \notin \Phi$. By Proposition 4.1, $\dim L^{d-2} > \dim L^2$, so the Hilbert function of L is not symmetric. □

9. Further remarks and questions

9.1. **The size of the defect in degree two.** The proof shows $\dim L^{d-2} \geq \dim L^2 + 1$. In all computed cases ($n \leq 8$) the defect is in fact given by the following formula.

Question 9.1. For $m \geq 3$, is it true that $\dim L^{d-2} = \dim H^4(X_{\mathbf{a}})$? Equivalently, do the degree-$(d-2)$ monomials in divisor classes span $H^{2(d-2)}(X_{\mathbf{a}})$?

Since $\dim H^4 = \binom{m}{2} + b - 1$ for $b \geq 1$ (and $\binom{m}{2} - 1$ for $b = 0$), while $\dim L^2 = \binom{m}{2} - [b=0]$, an affirmative answer would give $\dim L^{d-2} - \dim L^2 = b - 1 + [b=0]$. We verified this for all compositions with $m \geq 3$ and $n \leq 8$.

9.2. **Other types.** The same question makes sense for G/P with G of type B, C, D or exceptional. The argument above uses the product formula for the volume of coadjoint orbits, which holds in all types: $V(\mu) \propto \prod_{\alpha>0,\,\langle\alpha,\mu\rangle\neq 0}\langle\alpha,\mu\rangle$. It also uses Borel's presentation, so it should adapt. It would be interesting to determine whether the answer is again governed by the Levi factor, for instance whether symmetry holds exactly when P is maximal or the derived subgroup of the Levi factor is simple or trivial.

Appendix A. Computational verification

Hilbert functions. For each composition $\mathbf{a}$ with $n \leq 8$ we computed the kernel of the map $\mathbb{Q}[y_1,\dots,y_m] \to H^*(X_{\mathbf{a}})$ by elimination in Macaulay2 [9]: we eliminated x from the ideal generated by $e_1(x),\dots,e_n(x)$ and $y_j - \sum_{i\in B_j} x_i$, and read off the Hilbert function of the quotient. The resulting classification (Table 1) agrees with Theorem 1.1. Typical outputs:

$$\begin{aligned}
&(2,1,2): && (1,2,3,4,5,5,4,2,1),\\
&(2,2,2): && (1,2,3,4,5,6,7,8,9,7,5,2,1),\\
&(3,2,1): && (1,2,3,4,5,6,6,6,6,4,2,1),\\
&(2,1,1,1): && \text{symmetric } (L = H^*).
\end{aligned}$$

Consistency check of Proposition 5.2. For $\mathbf{a} = (2,2,1)$ we computed all intersection numbers $\int_X y^\beta c$, $|\beta| = d-2 = 6$, $c \in \{y_iy_j\} \cup \{p_2(B_k)\}$, exactly in the Borel presentation. We found $P_{y_iy_j} = \frac{1}{288}\partial_i\partial_j V$ and

$$P_{p_2(B_k)} = \tfrac{1}{288}\big(6\,V(g_k^2 - h_k) - \partial_k^2 V\big) \qquad (k=1,2),$$

in agreement with the form predicted by Proposition 5.2.

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